

Q9: Information Available in a Discrete Random Variable

Problem Statement

A factory inspects a product and records its quality using two different coding systems.

Each product falls into exactly one of the following four categories:

- G_1 : good quality, produced on Machine 1
- G_2 : good quality, produced on Machine 2
- D_1 : defective, produced on Machine 1
- D_2 : defective, produced on Machine 2

The sample space is:

$$\Omega = \{G_1, G_2, D_1, D_2\},$$

with probabilities:

$$P(\{G_1\}) = 0.50, \quad P(\{G_2\}) = 0.20, \quad P(\{D_1\}) = 0.10, \quad P(\{D_2\}) = 0.20.$$

Define two discrete random variables:

- X records only whether the item is good or defective:

$$X(G_1) = 0, \quad X(G_2) = 0, \quad X(D_1) = 1, \quad X(D_2) = 1.$$

- Y records both quality and machine:

$$Y(G_1) = 1, \quad Y(G_2) = 2, \quad Y(D_1) = 3, \quad Y(D_2) = 4.$$

Solutions

1. Probability space

$$\Omega = \{G_1, G_2, D_1, D_2\}, \quad \mathcal{F} = \mathcal{P}(\Omega), \quad P(\{\omega\}) \text{ as given.}$$

2. σ -algebra generated by X

Pre-images:

$$X^{-1}(\{0\}) = \{G_1, G_2\}, \quad X^{-1}(\{1\}) = \{D_1, D_2\}.$$

Thus:

$$\sigma(X) = \{\emptyset, \Omega, \{G_1, G_2\}, \{D_1, D_2\}\}.$$

3. σ -algebra generated by Y

Since Y is injective (all values distinct):

$$\sigma(Y) = \mathcal{P}(\Omega).$$

4. Inclusion $\sigma(X) \subseteq \sigma(Y)$

Clearly $\sigma(X) \subset \sigma(Y)$. Y contains more information because it distinguishes both quality and machine, while X only distinguishes quality.

5. Marginal distribution of X

$$P(X = 0) = P(G_1) + P(G_2) = 0.50 + 0.20 = 0.70$$

$$P(X = 1) = P(D_1) + P(D_2) = 0.10 + 0.20 = 0.30$$

6. Marginal distribution of Y

$$P(Y = 1) = 0.50, \quad P(Y = 2) = 0.20, \quad P(Y = 3) = 0.10, \quad P(Y = 4) = 0.20$$

7. Joint probabilities

$$P(X = 0, Y = 1) = P(G_1) = 0.50$$

$$P(X = 0, Y = 2) = P(G_2) = 0.20$$

$$P(X = 1, Y = 3) = P(D_1) = 0.10$$

$$P(X = 1, Y = 4) = P(D_2) = 0.20$$

8. Conditional probabilities

$$P(X = 1 \mid Y = 3) = \frac{0.10}{0.10} = 1$$

$$P(X = 1 \mid Y = 4) = \frac{0.20}{0.20} = 1$$

$$P(Y = 3 \mid X = 1) = \frac{0.10}{0.30} = \frac{1}{3}$$

$$P(Y = 4 \mid X = 1) = \frac{0.20}{0.30} = \frac{2}{3}$$

9. Interpretation

Conditioning on X tells only quality (good/defective), but not which machine. Conditioning on Y tells both exactly, removing all uncertainty.

10. Information comparison

$\sigma(X)$ has no information not in $\sigma(Y)$. $\sigma(Y)$ contains information about which machine produced the product, which $\sigma(X)$ lacks.

11. Python visualization

```
import matplotlib.pyplot as plt

outcomes = ['G1', 'G2', 'D1', 'D2']
probs = [0.50, 0.20, 0.10, 0.20]

plt.bar(outcomes, probs)
plt.ylabel('Probability')
plt.title('Probability Mass Function over ')
plt.show()
```

12. Shannon entropy

Marginal entropy of X :

$$H(X) = -[0.70 \log_2 0.70 + 0.30 \log_2 0.30] \approx 0.8813 \text{ bits}$$

Marginal entropy of Y :

$$H(Y) = -[0.50 \log_2 0.50 + 0.20 \log_2 0.20 + 0.10 \log_2 0.10 + 0.20 \log_2 0.20] \approx 1.761 \text{ bits}$$

Joint entropy $H(X, Y) = H(Y) \approx 1.761$ bits (since Y determines X).

Conditional entropies:

$$H(X|Y) = 0 \quad (\text{since } X \text{ is function of } Y)$$

$$H(Y|X) = H(Y) - H(X) \approx 1.761 - 0.8813 = 0.8797 \text{ bits}$$

Final Answers Summary

$$P(X = 0) = 0.70, P(X = 1) = 0.30$$

$$P(Y = 1) = 0.50, P(Y = 2) = 0.20, P(Y = 3) = 0.10, P(Y = 4) = 0.20$$

$$H(X) \approx 0.8813 \text{ bits}, H(Y) \approx 1.761 \text{ bits}, H(Y|X) \approx 0.8797 \text{ bits}, H(X|Y) = 0$$